

Week 3 : Assignment 3

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-08-12, 22:04 IST.

1. Which of the following statements is true about kernel and memory (RAM) in linux and xv6?

- ☒ Kernel resides in the upper part of the memory
- ☐ Kernel resides in the lower part of the memory
- ☐ Kernel does not reside completely in the memory
- ☐ Kernel resides anywhere in memory and this is kept track by a universal page table

No, the answer is incorrect. Score: 0

Accepted answer is: Kernel resides in the lower part of the memory

2. State True or false: There is a Kernel stack and a User stack in the process address space to ensure that the attacks that target the stack will not affect the kernel.

- ☒ True
- ☐ False

Yes, the answer is correct. Score: 1

Accepted answer is: True

3. A process after running for a while is preempted from the processor for other processes to run. Then its process state is moved to ____

- ☐ NEW
- ☒ READY
- ☐ RUNNING
- ☐ BLOCK

Yes, the answer is correct. Score: 1

Accepted answer is: READY

4. What is the output in the child process?

```
int main()
{
    char fg = '0';
    int pid = fork();
    if(pid != 0)
    {
        printf("fg = %d", fg+2);
    }
    else
    {
        printf("fg = %d", fg-2);
    }
}
```

- ☐ fg = M
- ☐ fg = Q
- ☒ fg = 77
- ☐ fg = 81

Yes, the answer is correct. Score: 1

Accepted answer is: fg = 77

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Yes, the answer is correct. Score: 1

Accepted answer is fg = 77

5. Which of the following are true about fork() system call (Multiple answers are correct)?

- ☒ All parent pages are initially marked as shared
- ☐ fork() returns 0 to the parent process
- ☐ fork() returns parent's process id to the child process
- ☒ To execute another program in the child process, another system call exec is evoked in the child process after fork

Yes, the answer is correct. Score: 1

Accepted answer is All parent pages are initially marked as shared, To execute another program in the child process, another system call exec is evoked in the child process after fork

6. Which of the methods can we use to terminate a process involuntarily?

- ☐ fork
- ☒ Kill
- ☐ exec
- ☐ exit

Yes, the answer is correct. Score: 1

Accepted answer is Kill

7. Which of the following are true about zombie entries?

- ☒ A zombie entry is removed when its parent reads its exit status
- ☐ When the process terminates, it becomes a zombie and is immediately removed
- ☐ All zombies are a resource leak and will continue to exist infinitely
- ☐ If a process terminates creating at least one intentional orphan, it does not become a zombie

Yes, the answer is correct. Score: 1

Accepted answer is A zombie entry is removed when its parent reads its exit status

8. State True or False: When an exit system call gets executed, a wakeup signal is sent to its parent process because the parent may be sleeping due to a wait.

- ☒ True
- ☐ False

Yes, the answer is correct. Score: 1

Accepted answer is True

9. State True or False: When the wait system call is invoked in the parent, it's going to be blocked until all of its child processes exit.

- ☐ True
- ☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

10. Which of the following is not part of an ELF header?

- ☐ Magic number
- ☐ Type
- ☒ Network details
- ☐ Machine details

Yes, the answer is correct. Score: 1

Accepted answer is Network details